COLOGNE | PhD application

Zhiheng Zhang

REVIEW COPY - NOT FOR SUBMISSION

Prepared 3 November 2026

Review checklist

1. Master's academic record: the supplied CSV and images are supporting records, not a verified official transcript. Confirm grades, combined research-project marks and pending results.

2. Thesis abstract: descriptive working title and draft text require applicant confirmation. No final results are claimed.

3. Publication status: reproduced from the existing application materials; confirm the current status before sending.

4. Undergraduate record: the newly supplied final-named transcript still states "UBC Credentials: None to date". The supplied graduation certificate is included separately for review; confirm which official record to submit.

Contents

CV; undergraduate transcript and graduation certificate; master's record summary and supporting images; draft thesis abstract; preprints and manuscripts.

This is a consolidated inspection copy. Review notices must be resolved before preparing the final single-PDF application. No email has been sent.

Zhiheng Zhang

zhiheng0mera@gmail.com

Academic Curriculum Vitae | PhD Applications, Fall 2027

Research Interests

Reliable intelligent software systems; multi-agent orchestration; software testing and program analysis; persistent agent execution, failure recovery, and evidence-based evaluation. Wearable and ubiquitous computing as application settings.

Education

University of Melbourne, Australia

Master of Computer Science | 2025-present; expected completion December 2026

Two-year program: one year of coursework followed by one year of research. Research supervisor: Professor Richard Sinnott. Master's research concerns Android privacy evidence; final results and degree completion are pending.

University of British Columbia, Okanagan campus, Canada

Bachelor of Science in Computer Science | 2020-2024; degree awarded 2024

Capstone Software Engineering Project (COSC 499): 93/100.

Preprints and Manuscripts

1. Zhiheng Zhang. Omission-Safe Conditional Contracts for Evidence-Closed CI Qualification. 2026. arXiv preprint, arXiv:2610.39241 [cs.SE].
Developed source-derived CI qualification contracts and an independent certificate checker to prevent missing observations from silently weakening evidence requirements. Research based on DS-Hns.

2. Zhiheng Zhang. Architectural Debt Enforcement under Incomplete Observation: A Controlled Study of Ratchets and Evidence Qualification. 2026. Manuscript submitted to arXiv; awaiting moderation.
Evaluated evidence-qualified architectural debt enforcement on 240 source files and 480 constructed scenarios, examining identity-aware ratchets, observation uncertainty, and provenance binding. Research based on Codex Boss (Boss-A2).

Research and Systems Experience

Utopia | Independent systems project

- Led architecture, AI-assisted implementation and integration, and evidence-based acceptance of a system combining persistent task execution with multi-agent orchestration.
- Integrated two physical Windows execution workers with Android and web control interfaces, explicit target-device routing, and inspectable cross-host results.
- Maintained failure histories, task-to-evidence links, and replay records to support debugging and reproducible evaluation.

- Completed and froze the integrated successor to DS-Hns and Codex Boss.

Privacy Lens | Master's research

- Worked on an offline Android privacy-evidence prototype and a curated reproducibility artifact under Professor Richard Sinnott's supervision.
- Organized evidence admission, metric checks, and claim-to-evidence traceability, distinguishing observations from interpretations that require human review.
- Worked with a TypeScript/React Native/Expo application and a Kotlin native layer.

DS-Hns | Persistent coding-task execution

- Designed durable checkpoints and resume checks around workspace state, repository identity, and unfinished effects before interrupted tasks continue.
- Used controlled interruption scenarios to distinguish committed-progress recovery from effect/checkpoint gaps that can still cause duplicate actions.
- Completed and froze the runtime; subsequently developed the CI evidence-qualification study listed above.

Codex Boss | Multi-agent orchestration

- Developed workflow orchestration around persistent task contracts, bounded recovery, and deterministic acceptance checks.
- Separated model-reported completion from host-observed evidence; subsequently integrated this work into Utopia.
- Completed and froze the platform; subsequently studied architectural debt enforcement and observation uncertainty in Boss-A2.

Technical and Research Practice

- Systems architecture: task lifecycles, interface contracts, durable state, failure handling, and multi-device integration.
- Evaluation: reproducible task scenarios, controlled fault injection, trace inspection, recovery/replay analysis, and evidence-backed acceptance.
- Software practice: JavaScript/Node.js, TypeScript, React Native/Expo, Kotlin/Jetpack Compose, Git/GitHub, and AI-assisted implementation with explicit review.
- Research development: problem formulation, experiment design, artifact organization, and technical documentation.

Selected Academic Preparation

Software Engineering; Capstone Software Engineering Project; Research Methods; Analysis of Algorithms.

Proposed Doctoral Focus

For doctoral research with the Trustworthy Artificial Intelligence group at Cologne, I am interested in testing and analyzing interacting AI agents to determine whether they satisfy intended outcomes under interruption, retries, and changing execution conditions. Interested in combining empirical failure evidence with program analysis and runtime checks, with the specific research problem refined in collaboration with a supervisor.

THE UNIVERSITY OF BRITISH COLUMBIA

TRANSCRIPT OF ACADEMIC RECORD

ENROLMENT SERVICES
2016 - 1874 East Mall
Vancouver, BC Canada V6T 1Z1

Surname: Zhang
Given Names: Zhiheng
Student Number: 54569462
Date of Birth: October 19

Date Printed: August 28, 2024

Issued To:
Zhang, Zhiheng
nicholas_zhang2020@163.com

REGISTRAR

Note: Transcript valid only if bearing the registrar's signature. This transcript is printed on secure paper on a blue background.

UBC Credentials

None to date

Transfer Credits

None to date

Winter Session 2020 - 2021

Bachelor of Science (UBC Okanagan) - **Year 1**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 111	(3.0)	Computer Programming I	73	B	3.0				389	71
1	COSC 122	(3.0)	Computer Fluency	93	A+	3.0				199	82
1	ENGL 150	(3.0)	Introduction to Literary Genre				W	2020/11/11			
1	MATH 100	(3.0)	Differential Calculus with Applications to Physical Sciences and Engineering	94	A+	3.0				357	69
1	PHYS 111	(3.0)	Introductory Physics for the Physical Sciences I	91	A+	3.0				274	80
2	COSC 121	(3.0)	Computer Programming II	64	C+	3.0				264	74
2	COSC 123	(3.0)	Computer Creativity	93	A+	3.0				124	86
2	ENGL 112	(3.0)	Studies in Composition	66	C+	3.0				27	75
2	MATH 101	(3.0)	Integral Calculus with Applications to Physical Sciences and Engineering	65	C+	3.0				341	75
2	MGMT 100	(3.0)	Introduction to Business	66	C+	3.0				136	74

Summer Session 2021

Bachelor of Science (UBC Okanagan) - **Year 1**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	MATH 221	(3.0)	Matrix Algebra	82	A-	3.0				38	68

Winter Session 2021 - 2022

Bachelor of Science (UBC Okanagan) - **Year 2**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 221	(3.0)	Introduction to Discrete Structures	77	B+	3.0				100	73
1	COSC 222	(3.0)	Data Structures	73	B	3.0				194	78
1	GEOG 108	(3.0)	Earth Systems: Weather, Climate, and Life	80	A-	3.0				167	76
1	MATH 200	(3.0)	Calculus III	77	B+	3.0				48	67
2	COSC 211	(3.0)	Machine Architecture	70	B-	3.0				129	67
2	ENGL 203	(3.0)	Topics in Composition	68	B-	3.0				20	85
2	MATH 225	(3.0)	Introduction to Differential Equations	63	C	3.0				48	64
2	STAT 230	(3.0)	Introductory Statistics	72	B	3.0				131	74

Summer Session 2022

Bachelor of Science (UBC Okanagan) - **Year 2**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 322	(3.0)	Introduction to Artificial Intelligence	79	B+	3.0				18	77
1	JPST 100	(3.0)	Beginning Japanese Language I	92	A+	3.0				26	77
2	STAT 124	(3.0)	Business Statistics	87	A	3.0				17	66

Winter Session 2022 - 2023

Surname:	**Given Names:**	**Student Number:**	**Date of Birth:**
Zhang	Zhiheng	54569462	October 19

Date Printed:
August 28, 2024

Winter Session 2022 - 2023 continued...

Bachelor of Science (UBC Okanagan) - **Year 3**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 301	(3.0)	Introduction to Data Analytics	86	A	3.0				144	77
1	COSC 304	(3.0)	Introduction to Databases	45	F	0.0				208	74
1	COSC 310	(3.0)	Software Engineering	86	A	3.0				109	86
1	MATH 302	(3.0)	Introduction to Probability	66	C+	3.0				11	60
2	COSC 303	(3.0)	Numerical Analysis	85	A	3.0				33	81
2	COSC 320	(3.0)	Analysis of Algorithms	82	A-	3.0				145	80
2	COSC 328	(3.0)	Introduction to Networks	62	C	3.0				79	75
2	GEOG 128	(3.0)	Human Geography: Space, Place, and Community	74	B	3.0				216	78
2	JPST 101	(3.0)	Beginning Japanese Language II	82	A-	3.0				21	74

Summer Session 2023

Bachelor of Science (UBC Okanagan) - **Year 3**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	INDG 100	(3.0)	Introduction to Decolonization: Indigenous Studies	68	B-	3.0				80	72

Winter Session 2023 - 2024

Bachelor of Science (UBC Okanagan) - **Year 4**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1	COSC 304	(3.0)	Introduction to Databases	90	A+	3.0				231	79
1	COSC 341	(3.0)	Human Computer Interaction	73	B	3.0				144	83
1	COSC 344	(3.0)	Image Processing and Applications	88	A	3.0				77	82
1	DATA 101	(3.0)	Making Predictions with Data	79	B+	3.0				107	71
2	COSC 414	(3.0)	Computer Graphics	76	B+	3.0				93	73
2	HEAL 100	(3.0)	Introduction and Principles of Health and Wellbeing	79	B+	3.0				294	84
2	MDST 120	(3.0)	Introduction to Computational Art and Design II	84	A-	3.0				26	86
2	PHIL 331	(3.0)	Computer Ethics	84	A-	3.0				79	92

Summer Session 2024

Bachelor of Science (UBC Okanagan) - **Year 4**

Term	Course	Credit Value	Course Title	% Grade	Letter Grade	Credit Rec'd	Stdg	Withdraw Date	Complete Date	Class Size	Class Avg
1-2	COSC 499	(6.0)	Capstone Software Engineering Project	93	A+	6.0				59	86

********** End of Record **********

THE UNIVERSITY OF BRITISH COLUMBIA

The University of British Columbia was established by an Act of the Provincial Legislature in 1908 and opened in 1915. The University is a member of the Universities Canada. UBC Okanagan was established as a campus of The University of British Columbia in April 2005. UBC Okanagan assumed responsibility for several Okanagan University College degree programs and all continuing students in those programs on July 1, 2005. Okanagan University College ceased its educational operations as of August 31, 2005.

EXPLANATION OF STUDENT ACADEMIC RECORD

Academic terms

The University operates on a year system. The academic year is divided into two sessions. Winter Session is divided into two terms, Term 1 from September to December and Term 2 from January to April. Summer Session starts in May and ends in August and is comprised of Term 1 (May and June) and Term 2 (July and August). Courses may span a single term (Term 1 or 2) or a full session (Term 1 and 2) for either session.

Distance Education and Technology (DET) courses begin on one of two starting dates during Winter Session, designated in the Term column as A,B,C or D. During the Summer Session, DET courses have only one starting date and are designated with an A or B in the Term column.

Academic Awards

All academic awards (Fellowships, Scholarships, Prizes and Medals) administered by the University and issued subsequent to April 1975, are noted on the transcript of academic record.

Course Numbering

Courses numbered 100 to 199 are primarily for the first year students. Similarly, courses numbered 200 to 299 are primarily for second year students, courses numbered 300 to 399 are for third year students, and courses numbered 400 to 499 are for fourth year students. Courses numbered 500 to 699 are considered graduate-level. Courses numbered 700 and above are residency courses.

Class Size and Class Average Grades

Effective February 1999, class size and class average grades are shown for most courses. In courses with more than one section, the class size and average are reported for the student's section.

Grading and Course Weight

In May of 1991, The University of British Columbia adopted a percentage grading and credit weighting system. Course weight is expressed in "credits". In general 1 credit represents 1 hour of instruction or 2 to 3 hours of laboratory work per week throughout one term of the Winter Session. A 1-unit course became a 2-credit course. Courses are normally graded on a percentage basis with a corresponding letter grade derived as shown.

Degree Standings and Averages

Standings are expressed as Class 1 (80% or over), Class 2 (65 to 79%), Class P (50 to 64%) or Honours; not used by all faculties. Effective Winter Session 1996/97, the calculation of the standing and average is based on the average percentage grade of all upper-level (300 or higher) courses used to satisfy the degree requirements (excluding courses graded as Pass/Fail); not used by all faculties. This is not applicable to programs in the Faculty of Graduate Studies and Postdoctoral Studies.

Undergraduate Grading Scale

Percent	Letter Grade	Percent	Letter Grade
90–100	A+	64–67*	C+
85–89	A	60–63*	C
80–84	A-	55–59*	C-
76–79	B+	50–54*	D
72–75	B	0–49	F* (Fail)
68–71	B-		

Graduate Grading Scale

Masters Grading Scale		Doctoral Grading Scale	
Percent	Letter Grade	Percent	Letter Grade
90-100	A+	90-100	A+
85–89	A	85–89	A
80–84	A-	80-84	A-
76–79	B+	76-79	B+
72–75	B	72-78	B
68–71	B-	68-71	B-
64–67	C+	0-67	F* (Fail)
60–63	C		
0-59	F* (Fail)		

*Some programs may require highter passing grades; failing grades are assigned a letter grade of "F"

The following standings also apply:

- AEG - aegrotat standing: indicates that the student is granted credit although unable to complete because of illness. A letter grade will also be recorded and that grade converted to a minimum percentage grade for that category for the calculation of averages.
- AUD - audit
- CH - challenge credit
- CIP - course in progress
- CR/D/F - (introduced in 2010/11) excluded from calculation in all averages; granted where a grade is assigned by an instructor but not included on the student's official transcript; grade is converted to CR, D or F standing. CR– credit, satisfactory completion of the requirements of the subject; D– marginal pass (50% – 54%), and not available for those programs or courses that define a passing grade as higher than 50%; F – fail
- EX - exempt
- J - see "A" standing as defined in gradeing scale prior to May 1991
- P or F - no grade assigned, excluded from calculation in all averages; P– pass, requirements of subject completed satisfactorily, credit granted where applicable; F – fail
- H - (introduced 1999/00 session) no grade assigned, excluded from calculation in all averages; H – honours, exceeds course require ments, credit granted where applicable
- PLA - credit assigned based on prior learning in subject area.
- S - see "S" standing as defined in grading scale prior to May 1991
- SD - standing deferred; excluded from calculation in all averages
- T - thesis in progress; graduating essay not submitted; course continuing
- W - official withdrawal

PRIOR TO MAY 1991

Course Weight and Hours

Course weight was expressed in "units". In general, 1 unit represented 1 lecture hour per week (or a laboratory period of approximately 2 to 3 hours each week) throughout both terms of the Winter Session. Some faculties did not assign an unit value to some courses. The maximum mark possible (max) indicated the weight of a course. A 3-unit course (3 hours of lectures per week throughout the Winter Session) was graded out of a total of 150 marks. This course approximated a 6 semester-hour or a 9 quarter-hour course.

Grading classifications

- Class 1 - 80% or better (i.e., 120 or more on a maximum of 150)
- Class 2 - 65% – 79% (i.e., 98 to 119 out of 150)
- Class P - 50% – 64% (i.e., 75 to 97 out of 150) except Dentistry, Graduate Studies, Librarianship, Medicine, Nursing and Rehabilitation Medicine where the range is 60% – 64%
- Class F - failure, no credit granted: below 50% except Dentistry, Graduate Studies, Librarianship, Medi cine, Nursing and Rehabilitation Medicine: below 60%

The following standings also apply:

- A - adjudicated pass: indicates that credit is granted and the course need not be repeated although it may not normally qualify as a prerequisite for further work. The grade as signed by the instructor is used in calculation of averages
- AEG - aegrotat standing: granted to a student who has been successful in studies but unable to write the final exam because of illness or other extenuating circumstances
- AUD - audit
- C - requirements of subject completed satisfactorily; no grade assigned; unit value granted where applicable; excluded from calculation in all averages
- D - standing deferred; excluded from calculation in all averages
- E - exempt
- N - no credit granted; did not write examination or otherwise complete requirements of the course (discontinued begin ning 1985– 86 session)
- S - supplemental privilege granted; failure; no credit granted; supplementals are not permitted in all faculties
- T - thesis in progress; graduating essay not submitted; course continuing
- W - (introduced in 1988/89 session) official withdrawal
- X - time expired; student did not complete course

Year Standing

In most Faculties each "year" was graded as Class 1, Class 2, Class P, Fail or carried a descriptive clause indicating action taken by the Adjudication Committee.

Degree Standing

Expressed as Class 1, Class 2, Class P, or with Honours; not used by all faculties. In some faculties the calculation of standing was based on the studies of the final two years, in others on the final year only.

THIS DOCUMENT CONTAINS SECURITY FEATURES. HOLD AT AN ANGLE TO VERIFY A WATERMARK BASKET WEAVE DESIGN

Undergraduate graduation certificate

Supplied document - review alongside the transcript.



The University of British Columbia

Kelowna, Canada

The Chancellor of The University of British Columbia
with the approval of the Okanagan Senate upon the recommendation of

The Faculty of Science

has conferred the degree of

Bachelor of Science

Major in Computer Science

on

Zhiheng Zhang

In Witness Whereof and by the authority duly committed
to us we have hereunto set our hand and seal
August 2024

Chancellor

President

Registrar

Dean

Master's academic record

UNOFFICIAL SUMMARY - PENDING CONFIRMATION

Source: supplied Master Score.csv. This table is not a University of Melbourne official transcript.

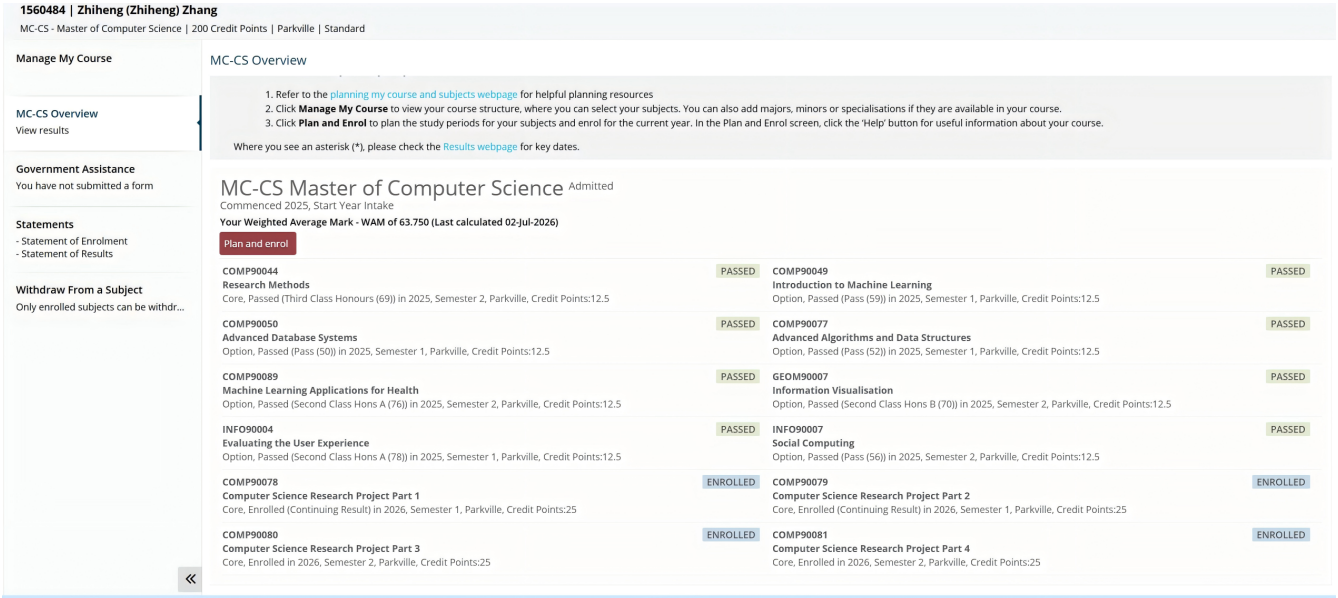
Year / term	Subject	Mark (%)	Credits	Status
2025 / Semester 1	COMP90049	59	12.5	completed
2025 / Semester 1	COMP90050	50	12.5	completed
2025 / Semester 1	COMP90077	52	12.5	completed
2025 / Semester 1	INFO90004	78	12.5	completed
2025 / Semester 2	COMP90044	69	12.5	completed
2025 / Semester 2	COMP90089	76	12.5	completed
2025 / Semester 2	INFO90007	56	12.5	completed
2025 / Semester 2	GEOM90007	70	12.5	completed
2026 / Semester 1	COMP90078	75	25	completed
2026 / Semester 1	COMP90079	75	25	completed
2026 / Semester 2	COMP90080	PENDING	25	pending
2026 / Semester 2	COMP90081	PENDING	25	pending

Derived WAM: 67.5/100 across 150 assessed credit points. This is not a final or officially certified WAM.

COMP90078 and COMP90079 share the combined 30/40 research-project result, normalized to 75% across 50 credits; they are not two independent 30/40 results. COMP90080 and COMP90081 remain pending. Confirm this treatment against the official transcript.

Master's supporting record

Master WAM.png | UNVERIFIED SUPPORTING IMAGE - NOT AN OFFICIAL TRANSCRIPT



Year	Study Period	Subject	Course Title	Mark	Credit Point	
2025	Semester 1	COMP 90049	Introduction to Machine Learning	59	12.5	Term Credit = 50,
2025	Semester 1	COMP 90050	Advanced database system	50	12.5	Cumulate Credit = 50
2025	Semester 1	COMP 90077	Advanced algorithm and data structure	52	12.5	Term WAM = (59+50+52+78)/4 = 59.75
2025	Semester 1	INFO 90004	Evaluating User Experience	78	12.5	Cumualte WAM = 59.75 * 50 / 50
						Cumulate WAM = 59.75
2025	Semester 2	COMP 90044	Rasarch Mohods	69	12.5	Term Credit = 50
2025	Semester 2	COMP 90089	Machine Learing Applicatons for Heath	76	12.5	Cumulate Credit = 100
2025	Semester 2	INFO 90007	Social Computing	56	12.5	Term WAM = (69+76+56+70)/4 = 67.75
2025	Semester 2	GEOM 90007	Information Visualization	70	12.5	Cumualte WAM = (59.75 * 50 + 67.75 * 50) / 100
						Cumulate WAM = 63.75
2026	Semester 1	COMP 90078	Computer Science Research Project (Part 1)	30	25	Term Credit = 50
2026	Semester 1	COMP 90079	Computer Science Research Project (Part 2)		25	Cumulate Credit = 150
			This 2 combines, hold 40/100 of Project(1+2+3+4)			Term WAM = 30/40 = 75
						Cumualte WAM = (59.75 * 50 + 67.75 * 50 + 75 * 50) / 100
						Cumulate WAM = 67.5
2026	Semester 2	COMP 90080	Computer Science Research Project (Part 3)	TBD	25	
2026	Semester 2	COMP 90081	Computer Science Research Project (Part 4)		25	
			This 2 combines, hold 40/100 of Project(1+2+3+4)			

Master's supporting record

Master-WAM.png | UNVERIFIED SUPPORTING IMAGE - NOT AN OFFICIAL TRANSCRIPT

Year	Study Period	Subject	Course Title	Mark	Credit Point	Extra Note
2025	Semester 1	COMP 90049	Introduction to Machine Learning	59	12.5	
2025	Semester 1	COMP 90050	Advanced database system	50	12.5	
2025	Semester 1	COMP 90077	Advanced algorithsm and data structure	52	12.5	
2025	Semester 1	INFO 90004	Evaluating User Experience	78	12.5	
2025	Semester 2	COMP 90044	Rasarch Mohods	69	12.5	
2025	Semester 2	COMP 90089	Machine Learing Applicatons for Heath	76	12.5	
2025	Semester 2	INFO 90007	Social Computing	56	12.5	
2025	Semester 2	GEOM 90007	Information Visualization	70	12.5	
2026	Semester 1	COMP 90078	Computer Science Research Project (Part 1+2 - Research Proposal)	30	25.0	This two combines
2026	Semester 1	COMP 90079	Computer Science Research Project (Part 1+2 - Oral Presentation)		25.0	40/100 of Project
2026	Semester 2	COMP 90080	Computer Science Research Project (Part 3)	TBD	25.0	This two combines
2026	Semester 2	COMP 90081	Computer Science Research Project (Part 4)		25.0	60/100 of Project

Master's supporting record

UniMelb Score Scale.jpg | UNVERIFIED SUPPORTING IMAGE - NOT AN OFFICIAL TRANSCRIPT

Grade	Mark	Description
H1	80%–100%	First Class Honours
H2A	75%–79%	Second Class Honours Division A
H2B	70%–74%	Second Class Honours Division B
H3	65%–69%	Third Class Honours
P	50%–64%	Pass
N	0–49%	Fail
NH	49%	Fail - you hadn't satisfactorily completed all prescribed (hurdle) requirements and would otherwise have passed the subject. A mark of 49% appears on your transcript. For more details, see our hurdle requirements FAQ
S	-	A special or supplementary exam has been awarded, and the exam will be held in the special and supplementary exam period. Special assessment is awarded as an outcome of Special Consideration.
WAF	-	Withheld – assessment to be finalised. This would be used, for example, when: • Special assessment (which will not be a special exam held during the special or supplementary exam period) has been awarded as an outcome of a special consideration application. • Other additional assessment is being considered. • The result has been withheld pending the outcome of an academic misconduct hearing. • Details will be provided via email.
WXT	-	An extension has been granted
CMP	-	Pass (no mark awarded). Only used for subjects marked on a pass/fail basis.

THESIS ABSTRACT - PENDING CONFIRMATION

Master's thesis abstract — draft for applicant verification

Author: Zhiheng Zhang
Program: Master of Computer Science, University of Melbourne
Supervisor: Professor Richard Sinnott
Working descriptive title: Evidence Traceability for Android Privacy Analysis
Status: Research in progress; official thesis title and final abstract require confirmation.

Android privacy analysis requires a careful distinction between observed behaviour and interpretations that the available evidence cannot establish. This master's research concerns Privacy Lens, an offline Android privacy-evidence prototype and a curated reproducibility artifact. The work organizes evidence admission, metric checks, and claim-to-evidence traceability to make the basis of privacy-related observations inspectable. The prototype uses a TypeScript/React Native/Expo application with a Kotlin native layer. Its research emphasis is on documenting what the collected evidence supports, preserving the boundary between observations and interpretations that require human review, and supporting reproducible inspection of the artifact. Final evaluation results and thesis conclusions remain pending and are not claimed in this draft.

Internal review note: Replace the working title with the registered title; confirm the scope against the thesis and add only verified methods and results before submission.

PUBLICATION STATUS - CONFIRM BEFORE SUBMISSION

Preprints and Manuscripts

Zhiheng Zhang

1. Zhiheng Zhang. Omission-Safe Conditional Contracts for Evidence-Closed CI Qualification. 2026. arXiv preprint, arXiv:2610.39241 [cs.SE].
2. Zhiheng Zhang. Architectural Debt Enforcement under Incomplete Observation: A Controlled Study of Ratchets and Evidence Qualification. 2026. Manuscript submitted to arXiv; awaiting moderation as recorded on 31 October 2026.

Neither item is presented as a peer-reviewed conference or journal publication. Status reproduced from the application repository's latest CV; recheck before submission.